SALMA: Structure-Aware Mask-Guided Alignment for Referring Image and Video Segmentation

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Abstract

Unified multimodal large language models (MLLMs) have shown strong potential for referring image and video segmentation, but their cross-modal attention can drift toward salient distractors when queries require fine-grained spatial reasoning. We propose **SALMA**, a structure-aware alignment method that injects an explicit mask prior into MLLM-based segmentation with minimal additional cost. SALMA performs a stop-gradient pre-pass through the SAM-2 mask decoder using a null prompt to obtain class-agnostic low-resolution mask logits. These logits are converted into a temperature-scaled spatial gate and used to modulate cross-attention outputs through a residual pathway with a learnable scale and warm-up schedule, preserving the pretrained model behavior during early optimization. We further introduce a text–mask contrastive objective and a boundary consistency loss to improve semantic and structural alignment. Experiments on image and video referring segmentation benchmarks show that SALMA improves over the Sa2VA-1B baseline, achieving 78.4 cIoU on RefCOCOg and gains of +3.4 and +1.7 J&F on Ref-DAVIS17 and Ref-YouTube-VOS, respectively, while adding only approximately 0.7% latency overhead. Code will be released upon acceptance.

1 Introduction

Referring segmentation—the task of localizing and delineating objects described by natural language—lies at the core of human–machine interaction. From embodied agents navigating cluttered environments ("pick up the red mug on the left") to assistive systems for visually impaired users and precision agriculture Luo et al. [2024], the ability to ground free-form language in pixel-precise masks is essential for actionable intelligence Lai et al. [2024], Rasheed et al. [2024]. Early specialist models Yang et al. [2022], Wang et al. [2022], Wu et al. [2022] achieved impressive results by designing task-specific architectures, yet they remain confined to task-specific benchmarks and lack the open-ended generalization demanded by real-world applications.

The recent surge of Multimodal Large Language Models (MLLMs) offers a promising alternative. By unifying visual perception with large-scale language reasoning, models like LLaVA Liu et al. [2024a], InternVL Chen et al. [2024], and Qwen-VL Bai et al. [2023] demonstrate remarkable zero-shot capabilities across diverse vision-language tasks. Building on this foundation, unified grounding frameworks such as Sa2VA Yuan et al. [2025] and LIRA Li et al. [2025] integrate LLM reasoning with the pixel-level precision of SAM-2 Ravi et al. [2024], aiming to enable general-purpose grounding within a single architecture.

However, a critical semantic-structural gap persists: existing MLLMs often treat visual features as abstract semantic tokens, weakening the low-level spatial priors needed for boundary delineation and instance-level disambiguation. This leads to *attention hallucination*—as shown in Figure 1, complex spatial queries (e.g., "*the cup behind the laptop*") can cause attention to drift toward salient distractors

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Figure 1: **The Attention Hallucination Problem in Unified MLLMs.** When faced with complex spatial queries (e.g., depth, ordering, interaction), existing unified MLLMs such as OMG-LLaVA Zhang et al. [2024b] (Center) exhibit severe attention drift, erroneously segmenting salient distractors rather than the queried target. This motivates our investigation into explicit spatial constraints. SALMA (Right) is our proposed solution.

rather than the spatially defined target. Our ablations in Table 2 indicate that explicitly constraining cross-modal attention is one of the most effective ways to reduce this failure mode.

To bridge this gap, we propose SALMA, a structure-aware method that re-introduces spatial fidelity into the attention mechanism. The key design choice is to reuse the SAM-2 decoder itself as a lightweight source of class-agnostic mask priors, rather than adding a separate localization head or relying on extra user prompts. By injecting these priors through Mask-Biased Attention inside the transformer layers, SALMA modulates cross-modal attention with dense structural cues, suppressing distractors while preserving the pretrained semantic representation.

Built upon this insight, our contributions can be summarized as follows:

- **Empirical Diagnosis: Spatial Constraint Deficiency.** We identify insufficient explicit spatial constraints in cross-modal attention as a major factor behind attention drift in unified MLLMs, supported by ablations and qualitative attention analysis.
- **Mask-Biased Attention (MBA).** We propose a soft residual attention-gating mechanism that repurposes the SAM-2 decoder through a stop-gradient pre-pass, turning the segmentation decoder into an active spatial guide without introducing an additional localization model.
- **Strong Performance with High Efficiency.** SALMA improves a Sa2VA-1B baseline across image and video referring segmentation benchmarks, achieving 71.9% J&F on Ref-DAVIS17 Pont-Tuset et al. [2017] and 78.4 cIoU on RefCOCOg Mao et al. [2016] with only a 0.7% latency increase.

2 Related Work

Large Multimodal Foundation Models. Large Language Models (LLMs) OpenAI [2023] have reshaped AI and motivated Multimodal LLMs that align visual encoders with language backbones. Early models such as LLaVA Liu et al. [2024a] focus on static images, while later work scales resolution Chen et al. [2024], Bai et al. [2023], strengthens reasoning Li et al. [2024b], and extends to video Lin et al. [2024], Bai et al. [2024]. Despite broad benchmarks Liu et al. [2024b], Fu et al. [2023], Li et al. [2024a], many MLLMs still struggle with structure because images are processed as coarse semantic tokens.

Visual Segmentation and Grounding. Before unified MLLMs, referring segmentation was driven by specialist models, often with modular language–vision designs. Transformer-based methods place attention at the core of fusion; LAVT Yang et al. [2022] and CRIS Wang et al. [2022] inject language early and use contrastive alignment. For video, ReferFormer Wu et al. [2022] treats RVOS as sequence prediction, while recent work strengthens grounding foundations Liang et al. [2025], global-local video reasoning Lin et al. [2025], Zheng et al. [2025], and interaction-aware expressions Jin et al. [2025]. In parallel, universal segmentation advanced rapidly (e.g., SEEM Zou et al. [2023], SegGPT Wang et al. [2023]), and SAM-2 Ravi et al. [2024] introduced promptable streaming memory. These models improve segmentation or temporal grounding, but they typically lack explicit LLM-level structural reasoning inside cross-modal attention.

Unified Multimodal Perception. To connect semantic reasoning with pixel grounding, recent work integrates segmentation decoders into MLLMs Lai et al. [2024], Ren et al. [2024], Rasheed et al. [2024], Xia et al. [2024], Yuan et al. [2025], Li et al. [2025], Wei et al. [2025], Wang et al. [2024]. LISA Lai et al. [2024] uses `[SEG]` tokens to invoke SAM-style decoding Kirillov et al. [2023], PixelLM Ren et al. [2024] adopts a lightweight pixel decoder and codebook, and GLaMM Rasheed et al. [2024]/PSALM Zhang et al. [2024c] add region- or mask-token interfaces. More recent systems broaden this line: InstructSeg Wei et al. [2025] unifies instructed image and video segmentation, SegLLM Wang et al. [2024] explores multi-round reasoning segmentation, Omni-RGPT Heo et al. [2025] studies region-level image/video understanding via token marks, VideoGLaMM Munasinghe et al. [2025] scales pixel-level video grounding, and UniPixel Liu et al. [2025] connects object referring with pixel-level visual reasoning. These designs are powerful, but most pass segmentation intent through sparse tokens, prompts, or implicit alignment objectives. SALMA addresses a complementary problem by imposing a dense structural prior inside attention: SAM-2 supplies class-agnostic candidate regions, and the MLLM resolves which candidate satisfies the referring expression. Building on this analysis, we next detail the SALMA methodology.

3 Methodology

3.1 Problem Formulation

We build SALMA upon **Sa2VA** Yuan et al. [2025], which integrates MLLM-based semantic reasoning with SAM-2 Ravi et al. [2024] segmentation decoding, providing a strong baseline with open-ended generalization and foundational mask quality.

However, Sa2VA exhibits a structural bottleneck: while it excels at high-level semantic recognition, it degrades on fine-grained spatial instructions in two ways. (i) **Coarse localization:** the implicit `[SEG]` token injection produces fragmented masks when objects share similar attributes. (ii) **Structural hallucination:** although visual tokens retain positional information, cross-modal attention lacks an explicit dense structural prior, causing attention to drift toward distractors under complex spatial queries (e.g., relative positioning or heavy occlusion), as shown in Figure 1. Simply scaling data or parameters cannot resolve this structural deficit; an explicit mechanism to re-introduce spatial priors is required, motivating our Mask-Biased Attention design.

3.2 Approach Overview

The overall pipeline of our proposed SALMA is illustrated in Figure 2. The approach consists of three integrated stages: (1) **Shared Visual Encoder:** A visual encoder and an LLM process the input video and text to extract multi-scale visual features and text embeddings. (2) **Mask-Biased Attention:** The proposed MBA mechanism injects pixel-level structural priors from the SAM-2 decoder into the cross-modal interaction to prevent attention drift. (3) **Fine-grained Alignment:** Finally, the SAM-2 decoder receives the text-conditioned features and predicts the target mask under a dual-constraint strategy involving TMC loss and Boundary Consistency loss.

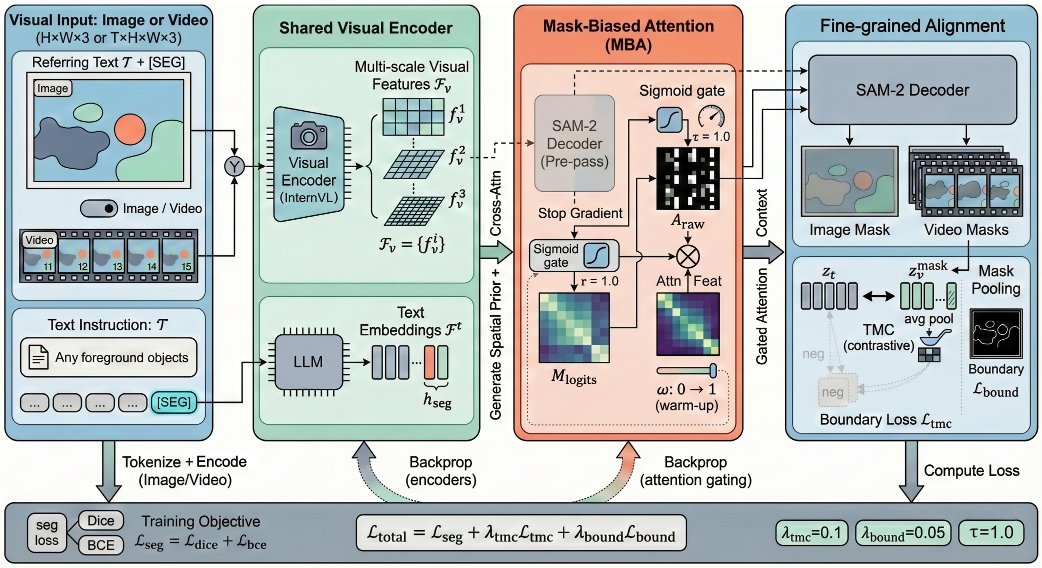

Figure 2: **Approach Overview. (a) Shared Visual Encoder.** Multi-scale visual features $\mathcal{F}_v$ are extracted by InternViT, while the LLM produces text embeddings $\mathcal{F}_t$ including the special `[SEG]` token h_{seg}. **(b) Mask-Biased Attention (MBA).** The *shared* SAM-2 decoder operates in a dual-mode fashion: a *stop-gradient pre-pass* (dashed arrows, stop-gradient) generates class-agnostic mask logits $\mathcal{M}_{logits}$, which are transformed via a temperature-scaled Sigmoid (τ_{gate}=1.0 in our final model) into a soft spatial gate G. This gate modulates ($\otimes$) the cross-attention output $\mathcal{O}_{attn}$ between visual queries and text keys/values, suppressing background noise. A deterministic linear warm-up coefficient $\omega(t)$ (rising from 0 to 1 over the first 5% of training steps) multiplies the learnable scale γ, so the effective gating strength is $\omega(t) \cdot \gamma$; this prevents the randomly initialized gate from disrupting pre-trained features in early training. **(c) Fine-grained Alignment.** The SAM-2 decoder, now with gradient tracking enabled, performs the final text-conditioned segmentation. Semantic alignment is enforced via TMC Loss ($\leftrightarrow$, contrastive matching between z_t and z_v^{mask}) and structural precision via Boundary Loss $\mathcal{L}_{bound}$ (Sobel edge matching). The total training objective combines $\mathcal{L}_{seg}$, $\mathcal{L}_{tmc}$, and $\mathcal{L}_{bound}$ with weights λ_{tmc}=0.1 and λ_{bound}=0.05.

3.3 Mask-Biased Attention

Standard cross-attention can introduce noise through unconstrained global interaction. We propose Mask-Biased Attention (MBA) to gate semantic injection using low-level spatial priors.

Spatial Prior via Auxiliary Execution. To obtain structural guidance without training a separate head, we leverage the *shared* SAM-2 decoder for a lightweight pre-pass. We invoke the decoder with a *null prompt* (i.e., no point, box, or mask input), relying entirely on SAM-2's learned internal saliency prior to produce class-agnostic mask logits. This design is motivated by three considerations: (i) the SAM-2 decoder, pre-trained on SA-V, already encodes strong saliency cues, making explicit prompting unnecessary; (ii) the null prompt is *constant* across training and inference, eliminating train–test distribution gaps; and (iii) the absence of user-supplied prompts keeps the prior purely data-driven. During pre-pass, output logits are detached from the computation graph; during final decoding, decoder weights are fine-tuned end-to-end. This dual-mode usage extracts robust priors without memory overhead.

Temporal Design for Video Inputs. For video referring segmentation, we adopt a *frame-independent priors with memory-consistent decoding* strategy. Specifically, the stop-gradient pre-pass generates spatial gates G independently for each frame, deliberately avoiding cross-frame propagation of the structural prior to prevent error accumulation from noisy early-frame predictions. Temporal coherence is instead handled by SAM-2's built-in memory attention mechanism during the *final decoding* stage (Stage 4 in Figure 2), where the decoder conditions on its memory bank to propagate identity-consistent masks across frames. This separation of concerns—frame-local structural gating for spatial

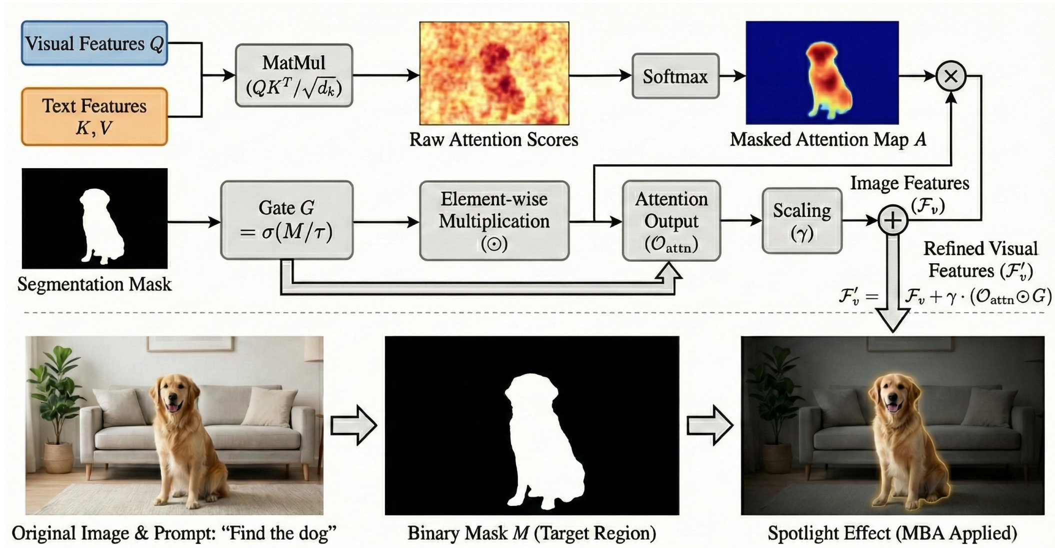

Figure 3: **The Mask-Biased Attention (MBA) Mechanism. Top:** Visual features $\mathcal{F}_v$ serve as Queries and text features as Keys/Values in cross-modal MHA. The resulting attention output $\mathcal{O}_{attn}$ is then element-wise multiplied ($\odot$) with a soft spatial gate $G = \sigma(\mathcal{M}/\tau)$. Crucially, this gate is derived from the *stop-gradient pre-pass* of the SAM-2 decoder, which provides class-agnostic structural priors without additional training. The gated features are injected back via a learnable residual: $\mathcal{F}_v' = \mathcal{F}_v + \gamma \cdot (\mathcal{O}_{attn} \odot G)$. **Bottom:** The resulting *spotlight effect*—a spatial concentration of attention onto the target extent—suppresses background hallucinations and confines semantic reasoning within the target's topology.

precision, memory-based propagation for temporal consistency—allows MBA to complement rather than interfere with SAM-2's native video capabilities.

Visual-Query Attention with Top-K Routing. Given the processed visual features $\mathcal{F}_v \in \mathbb{R}^{HW \times C}$ and text tokens $\mathcal{F}_t \in \mathbb{R}^{L \times C}$, we treat **visual features** $\mathcal{F}_v$ as **Queries** and **text tokens** $\mathcal{F}_t$ as **Keys/Values**, as illustrated in Figure 3. This design allows each spatial position on the dense visual grid to actively retrieve the most relevant semantic concepts from text, enabling fine-grained spatial-semantic alignment. To reduce computational redundancy, we employ a hard routing step that retains only a small subset of informative text tokens for cross-modal interaction. Specifically, we rank contextualized LLM hidden states by their detached ℓ_2 norm—a training-free heuristic that treats hidden-state magnitude as a proxy for token informativeness because high-norm token states tend to carry stronger residual activations after contextualization—and keep the top $K{=}2$ tokens for cross-modal interaction. Since this selection is discrete and parameter-free, gradients do not flow through routing; only the downstream cross-attention is optimized. We set $K{=}2$ as default; larger K yields diminishing returns. Let $\hat{\mathcal{F}}_t$ denote the gathered Top-K text tokens. For clarity, we present a simplified single-head form and omit per-head notation; the cross-modal attention map $A \in \mathbb{R}^{(HW) \times K}$ is computed as:

$$A = \text{Softmax}\left(\frac{(\mathcal{F}_v W_Q)(\hat{\mathcal{F}}_t W_K)^T}{\sqrt{d_k}}\right) \tag{1}$$

The retrieved semantic features $\mathcal{O}_{attn}$ are then computed via standard Multi-Head Attention (MHA) between the visual queries and the filtered text keys and values.

Mask-Gated Modulation. We apply gating to the cross-attention output rather than the internal probability map, modulating semantic flow without distorting the probabilistic alignment between tokens, as illustrated in Figure 3.

We first normalize the logits $\mathcal{M}_{logits}$ into a soft spatial gate G using a temperature-scaled Sigmoid; with $\tau_{gate}{=}1.0$, this reduces to a standard Sigmoid in our final model. The structure-aware features are then injected into the visual backbone via a gated residual connection:

$$G = \sigma\left(\frac{\mathcal{M}_{logits}}{\tau_{gate}}\right), \quad \mathcal{F}'_v = \mathcal{F}_v + \gamma \cdot (\mathcal{O}_{attn} \odot G) \tag{2}$$

where G is resized via bilinear interpolation to match the spatial resolution $H \times W$ of the visual query features $\mathcal{F}_v$. $\mathcal{O}_{attn}$ is the projected output of the Multi-Head Attention, $\odot$ denotes element-wise multiplication, and γ is a learnable scaling factor. To prevent the gate from perturbing pre-trained features before it has converged, we multiply γ by a deterministic warm-up coefficient $\omega(t)$ that linearly increases from 0 to 1 over the first 5% of training steps; the effective residual weight is therefore $\omega(t) \cdot \gamma$. After warm-up completes (ω=1), the network freely adapts γ via gradient descent. This learnable γ is crucial for robustness: it controls the global strength of the residual structural prior. If the pre-pass mask is noisy or incorrect (e.g., in low-contrast camouflage), the spatial gate G limits where the residual can act, while optimization can reduce γ to fall back toward standard attention. Empirically, on RefCOCOg the effective residual magnitude, measured as the spatial mean of γG, has a median of 0.38 but drops below 0.05 when the pre-pass mask covers $<$10% of the target, suggesting self-correcting behavior.

Soft Constraint vs. Hard Tokens. A key distinction of our approach is the nature of the spatial injection. Existing methods like LISA Lai et al. [2024] or GLaMM Rasheed et al. [2024] rely on "hard" tokens (e.g., `[SEG]`) to trigger segmentation, creating a representational bottleneck: the complex referential intent must be compressed into a single vector, often leading to "all-or-nothing" failures if the token embedding is misaligned. In contrast, MBA imposes a *soft, dense constraint* over the entire visual feature map. The residual connection $\mathcal{F}'_v = \mathcal{F}_v + \dots$ helps preserve the original semantic information rather than discarding it. If the prior is incorrect, the learnable γ can converge to zero, allowing the model to revert to standard attention—a robustness property difficult to achieve in hard-token architectures. Moreover, dense gating preserves fine-grained geometric cues for small or thin objects that vanish in token-based bottlenecks.

3.4 Fine-grained Alignment

MBA injects spatial priors, but precise masks still benefit from explicit fine-grained supervision. Without targeted constraints, features can drift to semantically similar yet spatially distinct instances (e.g., adjacent "red cars"). We therefore apply a dual-constraint strategy (TMC + boundary), as shown in the **Right** panel of Figure 2.

Text-Mask Contrastive Learning. To prevent the visual features from drifting away from the textual instructions during the decoding process, we enforce a semantic consistency constraint inspired by contrastive vision-language alignment Radford et al. [2021]. Specifically, we extract the region-level visual feature z_v^{mask} by average-pooling the post-MBA feature map $\mathcal{F}'_v$ over the ground-truth foreground region $\mathcal{M}^{gt}$. We use the ground-truth mask during training to promote stable semantic alignment, decoupling it from early-stage segmentation errors. Let z_t be the embedding of the special `[SEG]` token corresponding to the referring expression. We employ a Symmetric InfoNCE van den Oord et al. [2018] loss where *negatives are constructed from other text-image pairs within the same batch*, maximizing the mutual information between the matched text-visual pair (z_t, z_v^{mask}):

$$\mathcal{L}_{tmc} = \mathcal{L}_{v\to t} + \mathcal{L}_{t\to v} \tag{3}$$

where both directions (vision-to-text and text-to-vision) are normalized by a fixed temperature τ_{tmc}=0.15.

Boundary Consistency Constraint. Standard binary cross-entropy (BCE) and Dice losses primarily focus on the overall mask area but are insensitive to boundary errors. In video segmentation, however, fuzzy boundaries are a major source of qualitative degradation (e.g., temporal jitter). We incorporate a Boundary Loss $\mathcal{L}_{bound}$ that measures the discrepancy between predicted and ground-truth edge maps:

$$\mathcal{L}_{bound} = \|\widetilde{\nabla\hat{\mathcal{M}}} - \widetilde{\nabla\mathcal{M}^{gt}}\|_1 \tag{4}$$

where $\nabla\hat{\mathcal{M}}$ and $\nabla\mathcal{M}^{gt}$ denote the gradient magnitudes of the predicted and ground-truth masks, respectively, computed via Sobel edge detection with 3×3 kernels, and $\widetilde{\cdot}$ denotes per-sample min–max normalization applied to the edge maps for numerical stability. During training, $\mathcal{L}_{bound}$ is applied independently to every sampled frame (both image and video), ensuring that per-frame boundary precision is directly supervised rather than relying solely on temporal propagation. By optimizing this boundary-specific objective, the model is explicitly penalized for edge inaccuracies.

Table 1: **Comparison with Representative Unified MLLMs and Segmentation Methods.** We report cIoU for image referring segmentation tasks, $\mathcal{J}\&\mathcal{F}$ for video segmentation tasks, and standard metrics for multimodal understanding. Numbers of prior methods are from their original papers unless marked by *. The best reported results in each column are highlighted in **bold**. * denotes our reproduction under the same training data and hardware setup for fair comparison. A dash (–) indicates unsupported tasks or unreported results.

Method	Image Segmentation								Video Segmentation			Image Chat		
	RefCOCO			RefCOCO+			RefCOCOg		Ref-DAVIS17	Ref-YT-VOS	MeVis	MME	MMB	SEED
	Val	TestA	TestB	Val	TestA	TestB	Val	Test						
LLAVA-1.5-13B Liu et al. [2024a]	-	-	-	-	-	-	-	-	-	-	-	**1531(+)**	68.8	70.1
Video-LLaVA-7B Lin et al. [2024]	-	-	-	-	-	-	-	-	-	-	-	-	60.9	-
LLaMA-VID-7B Li et al. [2023]	-	-	-	-	-	-	-	-	-	-	-	1521(+)	65.1	59.9
mPLUG-Owl3-8B Ye et al. [2024]	-	-	-	-	-	-	-	-	-	-	-	-	77.6	-
InternVL2-8B Chen et al. [2024]	-	-	-	-	-	-	-	-	-	-	-	-	**81.7**	**76.2**
PixelLM-7B Ren et al. [2024]	73.0	-	-	66.3	-	-	69.3	-	-	-	-	309/135	17.4	-
PixelLM-13B Ren et al. [2024]	73.0	76.5	68.2	66.3	71.7	58.3	69.3	70.5	-	-	-	309/135	17.4	-
LaSagnA Wei et al. [2024]	76.8	-	-	66.4	-	-	70.6	-	-	-	-	0/0	0.0	-
LISA-7B Lai et al. [2024]	74.1	76.5	71.1	62.4	67.4	56.5	66.4	68.5	-	-	-	1/1	0.4	-
LISA-GLEE Wu et al. [2024]	76.4	78.2	73.8	67.3	71.3	62.3	71.6	72.4	-	-	-	-	-	-
GLaMM Rasheed et al. [2024]	79.5	83.2	76.9	72.6	78.7	64.6	74.2	74.9	-	-	-	14/9	36.8	-
LLaVA-G-7B Zhang et al. [2024a]	77.1	-	-	68.8	-	-	71.5	-	-	-	-	-	-	-
GSVA Xia et al. [2024]	77.2	78.9	73.5	65.9	69.6	59.8	72.7	73.3	-	-	-	-	-	-
SegLLM Wang et al. [2024]	80.2	81.5	75.4	70.3	73.0	62.5	72.6	73.6	-	-	-	-	-	-
OMG-LLaVA-8B Zhang et al. [2024b]	75.6	77.7	71.2	65.6	69.7	58.9	70.7	70.2	-	-	-	1177/235	47.9	56.5
PSALM Zhang et al. [2024c]	**83.6**	**84.7**	**81.6**	72.9	75.5	**70.1**	73.8	74.4	-	-	-	-	52.5	-
VideoLISA-3.8B Bai et al. [2024]	73.8	-	-	63.4	-	-	68.3	-	68.8	63.7	44.4	-	-	-
VISA-13B Yan et al. [2024]	72.4	-	-	59.8	-	-	65.5	-	70.4	63.0	44.5	-	-	-
LIRA-2B Li et al. [2025]	79.5	81.9	76.0	72.6	77.4	66.1	75.4	75.1	-	-	-	-	74.0	-
ReferDINO Liang et al. [2025]	-	-	-	-	-	-	-	-	68.9	**69.3**	-	-	-	-
Sa2VA-1B (Base)*	79.6	82.7	76.5	73.6	79.0	66.6	77.8	77.5	68.5	65.3	41.7	1487/433	71.9	71.0
SALMA	80.4	83.2	77.7	**74.8**	**80.0**	68.9	**78.0**	**78.4**	**71.9**	67.0	**46.0**	1421/353	71.2	69.9

3.5 Unified Training Objective

We formulate a unified training objective that balances global semantic fidelity with local structural precision:

$$\mathcal{L}_{total}(t) = \mathcal{L}_{seg} + \omega(t)\lambda_{tmc}\mathcal{L}_{tmc} + \omega(t)\lambda_{bound}\mathcal{L}_{bound} \tag{5}$$

where:

- $\mathcal{L}_{seg} = \mathcal{L}_{dice} + \mathcal{L}_{bce}$: This is the standard linear combination of Dice loss and Binary Cross-Entropy loss used in SAM-2, responsible for the general mask quality.
- $\mathcal{L}_{tmc}$: The Text-Mask Contrastive loss enforces semantic fidelity.
- $\mathcal{L}_{bound}$: The Boundary loss enforces high-frequency structural precision.

4 Experiments

4.1 Implementation Details

Architecture. We build on Sa2VA-1B, using InternVL2.5-1B Chen et al. [2025] (with InternViT encoder) as the MLLM backbone and SAM-2's Hiera-L as the segmentation image encoder. The SAM-2 mask decoder serves dual roles: generating detached logits during a stop-gradient pre-pass for prior extraction, and performing full-gradient text-conditioned decoding.

Training. We train end-to-end for 1 epoch on 4× RTX 5090 GPUs (per-GPU batch size 2, gradient accumulation 4) using AdamW (β_1=0.9, β_2=0.999, weight decay 0.05), learning rate 4×10^{-5} with cosine decay, DeepSpeed ZeRO-2, and BF16 mixed precision. The training mixture covers four data families: image referring segmentation (RefCOCO/+/g Kazemzadeh et al. [2014], Mao et al. [2016]), video referring segmentation (MeVis, ReVOS Yan et al. [2024], Ref-YouTube-VOS/Ref-DAVIS Xu et al. [2018], Pont-Tuset et al. [2017]), grounded conversation/region understanding (GCG Rasheed et al. [2024], Osprey-724k Yuan et al. [2024]), and general image/video instruction tuning (LLaVA-1.5 Liu et al. [2024a], Video-ChatUniVi Jin et al. [2024]). To keep structurally demanding segmentation examples visible during mixed training, we up-sample RefCOCO/+/g, MeVis, Ref-YouTube-VOS, Ref-DAVIS, and SAM-2 SA-V by $4\times$, ReVOS by $10\times$, and keep general instruction data at $1\times$; the full dataset table is provided in the supplementary material. Official train splits are used for all video benchmarks.

Hyper-parameters. We set λ_{tmc}=0.1, λ_{bound}=0.05, and τ_{gate}=1.0. A 5% linear warm-up (coefficient ω, see Section 3.3) is applied to auxiliary loss weights and the gating scale γ to prevent disrupting pre-trained features.

4.2 Evaluation Benchmarks

We evaluate across three complementary domains:

Referring Image Segmentation. We evaluate on the classic RefCOCO, RefCOCO+, and RefCOCOg benchmarks. These datasets test the model's ability to ground static objects from natural language descriptions. RefCOCOg is particularly challenging due to its longer, more complex expressions. We report performance using the standard Cumulative Intersection-over-Union metric.

Referring Video Segmentation. To assess structural stability in dynamic scenes, we test on Ref-DAVIS17 and Ref-YouTube-VOS. These benchmarks require the model to maintain consistent tracking of a referred object across video frames. We report the $\mathcal{J}\&\mathcal{F}$ mean score, which aggregates region similarity ($\mathcal{J}$) and boundary contour accuracy ($\mathcal{F}$).

Motion-Centric Understanding. We utilize the MeVis benchmark, which features queries explicitly dependent on motion cues (e.g., "the fish swimming to the left"). This tests whether our structural priors remain robust under complex temporal dynamics.

4.3 Comparison with Leading MLLMs

Quantitative Analysis. As shown in Table 1, SALMA improves over the strong Sa2VA-1B baseline on all evaluated referring segmentation benchmarks and remains competitive on general multimodal understanding metrics. Gains are largest in structurally demanding settings: +3.4 J&F on Ref-DAVIS17, +4.3 on MeVis, and +1.7 on Ref-YouTube-VOS, with 78.4 cIoU on RefCOCOg. Recent methods highlight the breadth–specialization trade-off: SegLLM Wang et al. [2024] improves single-round image referring segmentation while primarily targeting multi-round interaction, and ReferDINO Liang et al. [2025] achieves the highest Ref-YouTube-VOS score as a dedicated RVOS method. Larger recent systems such as InstructSeg Wei et al. [2025] are compared in the supplementary material due to their substantially larger model scale. SALMA remains competitive on video grounding, obtains the best Ref-DAVIS17 and MeVis scores among the listed unified MLLM-style models, and preserves general multimodal performance without relying on a video-specialist architecture. The stronger video gains suggest that accurate per-frame structural priors complement SAM-2's temporal memory, reducing the likelihood that early attention drift propagates across frames.

Generalization-Specialization Trade-off. We observe small decreases on general MLLM benchmarks (MME $-66/-80$, MMB -0.7, SEED -1.1), indicating that structure-centric adaptation mildly shifts model capacity toward grounding. This is consistent with our training recipe: the LoRA-adapted LLM is optimized with MBA, TMC, and boundary losses, and segmentation-heavy datasets are up-sampled for stronger spatial supervision. This suggests a controllable trade-off rather than an architectural failure: a less segmentation-heavy sampling mixture could further recover general chat performance when that objective is prioritized. Importantly, the decrease is limited compared with hard-token segmentation systems such as PSALM Zhang et al. [2024c], which reports 52.5 on MMBench versus 71.2 for SALMA, suggesting that soft residual gating better preserves general reasoning behavior.

4.4 Ablation Study

We investigate the separate contributions of each component in Table 2. The vanilla Sa2VA baseline exhibits substantial attention drift due to global cross-modal interaction without spatial constraints. Introducing MBA alone yields the largest single improvement (+1.31 J&F on DAVIS, +0.76 cIoU on RefCOCOg), suggesting that structural gating helps reduce attention hallucination. Adding TMC brings a further +1.21 on DAVIS but only +0.07 on RefCOCOg, indicating that contrastive alignment primarily benefits multi-frame disambiguation where semantic drift accumulates over time.

Table 2: **Ablation Study of Components.** We progressively add Mask-Biased Attention, Text-Mask Contrastive and Boundary Loss.

Configuration	DAVIS (J&F)	RefCOCOg (test)
Baseline (Sa2VA)	68.52	77.45
+ MBA	69.83	78.21
+ MBA + TMC	71.04	78.28
+ MBA + TMC + Bound	**71.87**	**78.42**

The Boundary loss adds +0.83/+0.14, sharpening edge precision. The full model achieves optimal performance (71.87 J&F / 78.42 cIoU), supporting our dual-constraint strategy.

Efficiency. We profile inference on 50 sampled Ref-DAVIS17 validation clips using a single RTX 5090 in BF16 precision. SALMA runs at 17.84 FPS versus 17.97 FPS for Sa2VA-1B, corresponding to a $\sim$0.7% latency increase. The additional cost is localized to the SAM-2 decoding stage: the stop-gradient pre-pass adds roughly 0.4 ms to the end-to-end per-frame latency, $<$1 GFLOP, and only +13 MB peak GPU memory.

Scaling and Additional Analysis. SALMA also scales to a stronger InternVL2.5-4B backbone, improving Sa2VA-4B on all RefCOCO/+/g splits with the largest gain on RefCOCOg Val (+0.9 cIoU), suggesting that dense structural gating remains useful beyond the 1B regime. Further ablations in the appendix show that replacing our spatial gate with Residual FiLM degrades Ref-DAVIS17 and MeVis, supporting the minimal-intervention design of MBA. We also include a stratified saliency analysis, full computational-cost breakdown, and qualitative visualizations in the appendix.

Limitations. SALMA is less effective on extreme micro-objects ($< 1\%$ area), where class-agnostic SAM-2 priors are unreliable. It can also fail in dense near-identical instances and abstract physical-state reasoning, where the spatial gate highlights plausible candidates but the MLLM must still resolve subtle semantic attributes. Future work may combine the spatial gate with active zoom-in or attribute-aware gating for such cases.

5 Conclusion

We identify the lack of explicit spatial constraints as a structural bottleneck in unified MLLMs for referring segmentation and introduce SALMA to address it. By repurposing the SAM-2 decoder as a class-agnostic prior generator, MBA provides dense spatial guidance with only $\sim$0.7% latency overhead. Across image and video referring segmentation benchmarks, SALMA improves over Sa2VA-1B, with ablations showing that explicit spatial gating contributes the largest single gain.

These results suggest that lightweight structural priors can improve foundation-model grounding without replacing pretrained semantic reasoning. Future work should make these priors more reliable for micro-objects and highly abstract referring expressions.

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A Additional Analyses and Qualitative Results

A.1 Feature Modulation Ablation

Table 3: **Analysis of Feature Modulation.** Comparing our simplified spatial gating vs. complex FiLM modulation. Red text indicates a performance drop relative to our full model.

Method	Ref-DAVIS17	MeVis (Val_u)
Ours (Full)	**71.87**	**53.37**
Ours (w/ FiLM)	68.13 (-3.74)	47.46 (-5.91)

We compared against Residual FiLM Perez et al. [2018], the canonical feature modulation paradigm ($\gamma \cdot \mathcal{F} + \beta$). Replacing our gating with FiLM substantially degrades performance, suggesting that full affine modulation distorts the pre-trained feature statistics used by SAM-2.

A.2 Saliency Bias Analysis

Table 4: **Saliency Bias Analysis on RefCOCOg (Val).** We report cIoU scores grouped cumulatively by object area ratio.

Method	$< 1\%$ ($N = 35$)	$< 2\%$ ($N = 151$)	$< 5\%$ ($N = 1559$)	All ($N = 4896$)
Baseline (Sa2VA)	**37.1**	47.7	68.0	77.8
Ours	33.2	**48.9**	**70.2**	**78.0**

SALMA improves on objects occupying $< 2\%$ and $< 5\%$ of the image area, while dropping on the extreme $< 1\%$ subset. This subset accounts for only 0.7% of the validation samples, but it identifies a useful failure mode for future high-resolution prompting.

A.3 Computational Cost

Table 5: **Computational Cost Analysis on 1024×1024 resolution.** The SALMA pre-pass introduces negligible overhead relative to the backbone encoders.

Component	Params (M)	FLOPs (G)
Vision Encoder (InternViT)	304.0	1614.1
SAM-2 Image Encoder (Hiera)	212.7	810.5
Visual Projector	4.5	1.2
SALMA Pre-Pass (Gating + Decoder)	12.0	$<$1.0

A.4 Qualitative Results

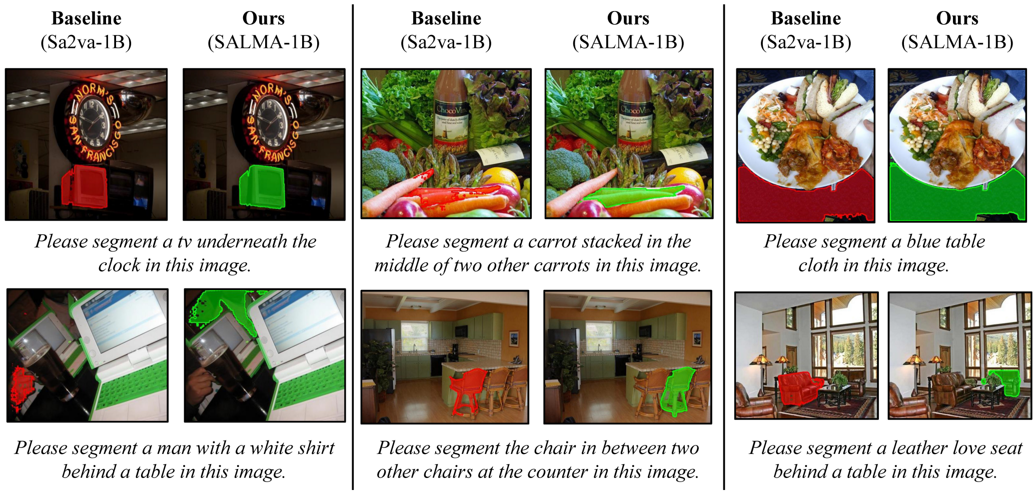

Figure 4: **Qualitative Comparison on Complex Referring Expressions.** We present six diverse examples comparing the Baseline (Sa2VA-1B) with SALMA.

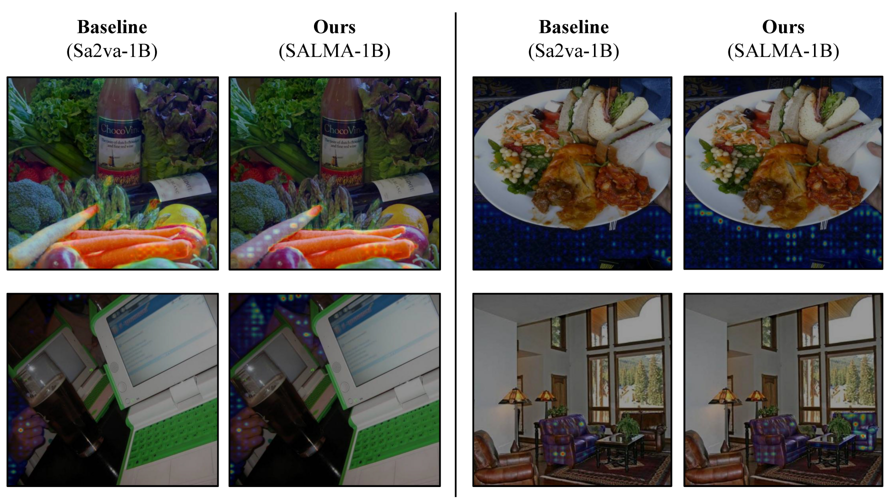

Figure 5: **Spatial gate maps (ours) vs. cross-modal attention (baseline).** Mask-Biased Attention helps mitigate attention drift under complex referring expressions.

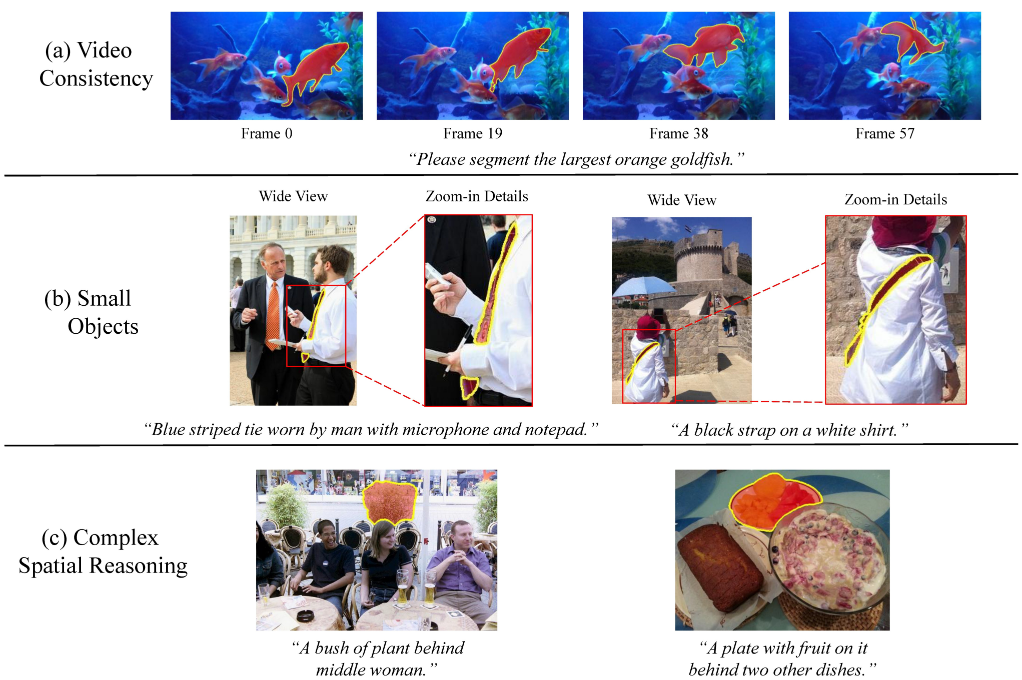

Figure 6: **Qualitative Results across Diverse Scenarios.** SALMA improves temporal consistency, small-object grounding, and complex spatial reasoning.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state that SALMA targets referring image and video segmentation by injecting structure-aware mask priors into cross-modal attention. The claims are supported by the main comparison table, ablations, and efficiency analysis.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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7. **Experiment statistical significance**

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   Justification: The paper follows the standard evaluation protocol for the reported segmentation and multimodal benchmarks, but it does not report multi-seed error bars or confidence intervals because training and evaluating these MLLM-based segmentation systems is computationally expensive.

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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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